A 212-variable quartic with additive height comparisons

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Abstract

We reduce the pinned 214-variable quartic family to 212 total variables, under the same explicitly retained arithmetic premises. The new family has 186 primitive quadratic equations and 2908 nonzero quartic monomials. Replacing the comparison arguments $\lambda, T\lambda$ by $\lambda, \lambda + T$ permits both contact calculations to use three rather than four outputs, even after the scalar normalization of the elliptic coordinates. A rational rescaling of two coordinates also retains the earlier coefficient-height improvement. The sufficient defining parameter remains ineffective. The changed arithmetic applications, all rational boundary cases, all retained rows, and all quartic coefficients are treated separately from the source's still incomplete independent arithmetic audit.

1 Statement and retained source

Every variable count includes the distinguished input. The source is the exact 214-coordinate, 188-row family at upstream commit `1f7e1f1fb0438feda3c9f3da9cc16982126aeeed`. Its numeric source, reconstructed from the pinned manuscript and checked against the published expanded quartic, is retained in

`computation/quartic/balanced214_v24/source`.

The source-row SHA-256 is

`49cf4066bf1e35e02e2156bf4bd24a21bb4c5e794f7374e056413d8306fb73c5`.

The coordinate list is separate from the quadratic table. Every new row and retained row is specified by the transformation in Section 2 and by the complete output table; no generic-coordinate deletion is being asserted.

The retained premises are the source elliptic group and index-height statements; its scalar extraction and prepared realization; its pole obstruction and prepared nonzero realization; the full-family geometric/modular contact-height supplier; the aggregate *sum* height theorem; and the fixed-data simultaneous-prime statement with arbitrary finite rational contact coefficients, finite avoidance, conjugate reduction orders, compatible common-index congruences and real neighborhoods. The exact supplying statements and their applications are recorded in the pinned source's `research/quartic214/proofs/uniform-existence.md`, P03–P20. The coefficient and field identities below do not certify those arithmetic foundations. No additional *abc* hypothesis is introduced.

Theorem 1.1. *Under these retained inputs, there is a computable family*

$$P_N \in \mathbb{Z}[X, Y_1, \ldots, Y_{211}], \qquad N \geq 1,$$

with exact degree four, content one, and 2908 nonzero coordinate monomials. There is one fixed integer N_0 such that every $N \geq N_0$ has rational input projection exactly $\mathbb{Z}$. No numerical sufficient N_0 is determined.

In the unbalanced presentation, put $A_N = 149^{2N+1}$. *Its exact height is* $\max(C_0, A_N^2 + 1)$, *where* C_0 *is the explicit constant in the coefficient certificate, and hence is* $149^{4N+2} + 1$ *for* $N \geq 53$. *In the balanced presentation put* $C = 149^N$ *and*

$$K_0 = 31581660^{28}, \qquad J_0 = 5898050446009651968, \qquad H_0 = (J_0 K_0)^2.$$

Its exact height is

$$\max(H_0, K_0 C^2),$$

and is $31581660^{28}149^{2N}$ *for* $N \geq 57$. *Both presentations have the same rational input projection at each fixed* N. *These coefficient thresholds are not defining thresholds, and the sufficient* N_0 *is not identified with the old 214 threshold.*

2 Exact changes to the quadratic equations

Use the pinned source coordinates before compaction. Its scalar normalizer is $R_s = x_{118}$, its square is $w = x_{117}$, and the root coordinates in four blocks are $C_i, E_i \in K = \mathbb{Q}(\sqrt{2})$. The unchanged scalar rows give

$$\beta = 51X^2 + 2504, \qquad r_s = \beta(\beta - 1)(\beta - 2401)(32\beta + 343) > 0,$$

$$(1 + 149^2 r_s)R_s = A_N r_s, \qquad w = R_s^2 > 0.$$

These identities use only retained rows. They can also be checked from source rows 48,49,60,63 and the input-square equation; the intermediate pair is $(51X^2 + 2501)^2/4$. The physical trace is

$$T = \frac{1}{4}\operatorname{Tr}\sum_{i=1}^{4} x([n_i]P), \qquad wT = \frac{1}{8}\sum_{i=1}^{4}\left((C_{i,0} - E_{i,0})^2 + 2(C_{i,1} - E_{i,1})^2\right). \tag{1}$$

Each root is a nonzero doubled point; its abscissa exceeds $3 \mp \sqrt{2}$ in the corresponding real embedding. Thus $T > 6$.

Repurpose coordinate 4 as $t_0 = \lambda$, and coordinate 90 as the *unscaled* second argument t_1. Replace source row 182 by

$$8w(t_1 - t_0) = \sum_{i=1}^{4}\left((C_{i,0} - E_{i,0})^2 + 2(C_{i,1} - E_{i,1})^2\right). \tag{2}$$

Since $w > 0$, this gives $t_1 = t_0 + T$. The equation remains quadratic; this is the point at which additive rather than multiplicative comparison permits removal of the old scaling on t_1.

Put

$$\Theta_0 = \{0, -1/2, -1, 1/2399, -32/407\}, \qquad \Theta = \Theta_0 \cup (2 - \Theta_0),$$

and

$$P_5(r) = \prod_{\theta \in \Theta_0}(r - (1 - \theta)^2) = r^5 - e_1 r^4 + e_2 r^3 - e_3 r^2 + e_4 r - e_5.$$

The five squared distances are $1, 9/4, 4, 5750404/5755201, 192721/165649$. For $j = 0, 1$, let q_j and s_j be the existing weight and its inverse; retain $q_j s_j = 1$. Replace the contact equations by four quadrics with three outputs r_j, u_j, v_j:

$$r_j = (t_j - 1)^2, \qquad u_j = r_j^2, \tag{3}$$

$$v_j = r_j u_j - e_1(1 + s_j)u_j + e_2(1 + s_j)r_j, \tag{4}$$

$$u_j v_j = (1 + s_j)(e_3 u_j - e_4 r_j + e_5). \tag{5}$$

The first block uses old slots $(76, 77, 78)$ and inverse weight 80. The second uses $(91, 92, 93)$ and inverse weight 95. Delete old coordinates 119 and 120 and their defining rows 64 and 65. The four equations replace source rows 22–25 and 35–38, respectively. Compact the remaining coordinates and rows in increasing old-index order. All other 177 rows are unchanged, up to this compaction and primitive normalization. In particular no local-factor exponent, point relation, scalar condition, or pole equation changes. There are $214 - 2 = 212$ coordinates and $188 - 2 = 186$ rows.

Lemma 2.1. *With $qs = 1$, equations (5) are equivalent to*

$$r = (t-1)^2, \qquad (1+q)P_5(r) = r^5.$$

The extra outputs are uniquely determined. Every solution has $P_5(r) \neq 0$. The boundary $t = 1$ is included and has $q = s = -1$.

Proof. Substitute $u = r^2$ and the expression for v into the last residual. The result is exactly

$$(1+s)P_5(r) - sr^5.$$

Multiplication by q gives the asserted weight equation. These steps are reversible because $q \neq 0$. If $P_5(r) = 0$, then $r = 0$, but $P_5(0) = -e_5 \neq 0$, a contradiction. At $r = 0$ the same equation forces $s = -1$ and then $q = -1$; the outputs are $u = v = 0$. □

Consequently t_j avoids all ten contacts, including 0 and 2, but $t_j = 1$ is not discarded. The first old contact representation is recovered, for purposes of comparison, by

$$\rho = 1/t_0, \quad \rho^2, \rho^4, \rho^5, \quad G(\rho) = t_0^{-10} P_5((t_0 - 1)^2).$$

The second old representation is recovered by

$$t_{1,\mathrm{old}} = wt_1, \quad r_{\mathrm{old}} = wr_1, \quad u_{\mathrm{old}} = wr_1^2, \quad v_{\mathrm{old}} = wr_1^3, \quad F_{\mathrm{old}} = w^2 P_5(r_1).$$

Every old contact equation is then satisfied exactly. This verifies the same physical weight q_j, not a replacement chosen only to have the right degree. The old aggregate equation is deliberately not retained; its new arithmetic use is proved next.

3 The changed height comparison

Let $M(t)$ be the odd contact radical outside the source fixed exceptional set Σ, comprising primes at most 76176 and 2029147. The source full-family height supplier, applied to the two reflected five-contact blocks, gives a fixed $H_2 > 0$ with

$$h(t) \leq H_2 M(t)^{7/6} \tag{6}$$

for every rational t away from the ten contacts. It includes $t = 1$, with radical one. This is the same source application as before: the contact set and excluded primes have not changed. The aggregate *sum* theorem supplies fixed $M_1 > 0, M_2 \geq 1$, with M_2 ineffective, such that

$$B_0^2 \leq M_1 h(T) + M_2, \qquad B_0 = \max_i |n_i|.$$

The index tuples and trace in (1) are the identical objects consumed by that theorem; no product-height statement is substituted.

For the new arguments the additive height inequality gives

$$h(T) = h(t_1 - t_0) \leq h(t_1) + h(t_0) + \log 2. \tag{7}$$

Thus whenever $M(t_j)^k \mid J$ separately for $j = 0, 1$, for one nonzero integer J, one has

$$B_0^2 \leq C|J|^{7/(6k)} + M_2', \qquad C = 2M_1 H_2, \quad M_2' = M_2 + M_1 \log 2. \tag{8}$$

All constants are independent of N, the input and every witness. Only a fixed additive constant changes. It is not necessary for the two radicals to be coprime.

4 Local implication and soundness

The unchanged algebraic rows recover the same physical reference, scalar and numerator points as in the source. The scalar normalizer is positive independently of the changed contact and aggregate rows. The source's forced-prime argument therefore excludes the forbidden reference torsion and zero denominators before logarithms, exactly as in P04 of its uniform-existence proof. The new contact lemma gives the same $q_j = R_j^{-1} - 1$ whenever $(t_j - 1) \neq 0$, with $R_j = P_5((t_j-1)^2)/(t_j-1)^{10}$; no division by $t_j - 1$ is used on the boundary or unless a contact prime is being consumed.

At a consumed prime, write $v(q_j) = -\nu$, with ν positive odd. The five unchanged physical factors are

$$\begin{aligned} 1 + q^5 C_2^2 + q^2 D_H^2 + q^4 B_1^4, \\ 1 + q^5 G^2 + q^2 A_3^2 + q^4 B_0^4, \\ 1 + q^5 I^4 + q^3 A_2^2 + q^2 A_4^2 + q^4 B_3^4, \\ 1 + q^5 C_3^2 + q^2 C_1^4 + q^4 B_4^4, \\ 1 + q^5 C_4^2 + q^2 A_1^2 + q^4 B_2^4. \end{aligned}$$

Here the notation is the source notation: D_H, A_i, B_i are reciprocal abscissas, C_i are differences of primary and conjugate formal ratios, I is the input comparison, and G is the Lucas residual in those ratios. Reciprocals have even valuation. The nonconstant summands have distinct valuation classes, including the special third factor, so negative valuations and zero tests cannot cause cancellation. The cleared trace and pole equations force the maximum place valuation of the factor product to exceed $-\nu/2$. They give order at least three for C_2, C_3, C_4, G, order at least one for C_1, I, and positive even reciprocal orders. Integrality of one full logarithmic ratio is transferred to the other before using relative order-four precision as an absolute congruence.

Consequently, for the same nonzero integer labels (n, a, t, b),

$$\Delta = t^2 - a(a-2)tb + (a-2)^2 b^2 - 1,$$

and for J the reduced numerator of $n - \beta$ when nonzero,

$$M(t_j)^3 \mid \Delta, \qquad M(t_j) \mid J \qquad (j = 0, 1). \tag{9}$$

The source proof of these conclusions uses the contact equation only at a consumed prime and uses neither $t_1/t_0 = T$ nor any other relation between the two arguments. Its hypotheses are therefore unchanged. The same bounds $|\Delta| \leq 14 B_0^4$ and $|J| \leq 2 B_0 e^{h(\beta)}$ apply.

The unchanged scalar conclusion is

$$c = |a| \geq 149^{L_N}, \qquad h(\beta) \leq 2^{330770} 149^{-L_N} c(1 + \log c/3), \quad L_N = \left\lfloor \frac{9(2N-1)}{4} \right\rfloor + 1. \tag{10}$$

If $\Delta \neq 0$, (8) gives $B_0^2 \leq C 14^{7/18} B_0^{14/9} + M_2'$, so B_0 is bounded by a fixed constant. The lower bound $B_0 \geq c \geq 149^{L_N}$ excludes this case for every sufficiently large fixed N, uniformly in the input and all solutions.

When $\Delta = 0$, put $c = |a|$ and $w_0 = \operatorname{sign}(a)(a-2)b$. Then $t^2 - ctw_0 + w_0^2 = 1$. The norm-one unit reduction with $\epsilon = (c + \sqrt{c^2 - 4})/2$ shows that $|w_0| = U_j$ and $|t| \in \{U_{j-1}, U_{j+1}\}$, where $U_0 = 0, U_1 = 1, U_{j+1} = cU_j - U_{j-1}$. For clarity, ϵ is the smallest norm-one unit greater than one in $\mathbb{Z}[\epsilon]$: any such unit with nonzero coefficient of ϵ differs from its inverse in absolute value by at least $\sqrt{c^2 - 4}$. Reduction into $[1, \epsilon)$ proves the assertion without assuming a maximal order. Since $w_0 \neq 0$ and $|a - 2| \mid w_0$, reduction of U_j modulo $c - 2$ or $c + 2$ gives $j \geq c - 2$. Induction gives $U_j \geq (c-1)^{j-1}$. For $c \geq 8$ this implies the convenient weaker bound

$$\log B_0 \geq (c-5)\log(c-2).$$

If also $J \neq 0$, (8) with $k = 1$ gives, once $B_0^2 \geq 2M_2'$,

$$\log B_0 \leq K + \tfrac{7}{5}h(\beta) \leq K + \epsilon_N c(1 + \log c/3), \qquad \epsilon_N = \tfrac{7}{5}\,2^{330770}149^{-L_N},$$

where K is fixed. Choose N large enough that $\epsilon_N \leq 1$ and $c \geq \max(e^{12}, 2\max(K,0) + 1, \sqrt{2M_2'} + 1)$. Then

$$(c-5)\log(c-2) - c - \tfrac{c}{3}\log c \geq c(\tfrac{1}{6}\log c - 1 - \tfrac{1}{2}\log 2) > c/2 > K,$$

a contradiction. Hence $n = \beta$ is integral. If $X = u/v$ is reduced, $v^2 \mid 51$; squarefreeness of 51 gives $v = 1$. This proves soundness with one fixed threshold. The additional $\log 2$ in (7) changes only that ineffective threshold, not the strict exponents.

5 Completeness with the new arguments

The scalar part is unchanged. Its source proof chooses a fixed threshold N_C independent of the integer input: for the positive integer scalar product r_s, one has $R_s \geq A_N/(1 + 149^2)$, and fixed compact real neighborhoods make the scalar pole input lie in $(0, 17)$ for every $N \geq N_C$. Finite-place depths may depend on (N, X), as before. Choose its positive scalar label a sufficiently divisible and large, then its nonzero Lucas labels t, b, and $n = \beta$. Their four points and the positive trace $T > 6$ are now fixed. No subsequent step imposes a further lower bound on N.

Choose fixed compact real and finite-place neighborhoods for the common reference point and all its numerator multiples, avoiding the source denominators. The new coefficient list in λ is

$$\Theta \ \cup \ (\Theta - T).$$

The span of Θ is 4 and $T > 6$, so there are exactly twenty distinct coefficients. The prime-selection premise permits this fixed list; it is not restricted to the nineteen or fewer values in a multiplicative list with a shared zero. Choose D_1 so that $D_1^2\theta$ and $D_1^2(\theta - T)$ are integers, include all fixed avoidance and preparation primes, and take $\lambda = u/(D_1^2 w_1^2)$. The selected prime numerators are

$$u - \theta D_1^2 w_1^2, \qquad u + (T - \theta)D_1^2 w_1^2 \quad (\theta \in \Theta).$$

All labels, coefficients and moduli are fixed before invoking the prime theorem. Its same conjugate annihilating orders and compatible common-index progression give the required primary/opposite formal groups while preserving the scalar data and real neighborhoods. At each selected contact, $v(q) = -1$, reciprocal tests have depth at least two, and comparison and Lucas tests have depth at least four. Every nonconstant summand of the five factors is then positive at the successful place. At the opposite place its inverse residue is zero or one. The reciprocal trace is a unit of residue one or two, exactly as in the source.

For either fixed offset $a_0 \in \{0, T\}$, with $v_0 = D_1^2 w_1^2$,

$$R_j = \frac{\prod_{\theta\in\Theta}(u + (a_0 - \theta)v_0)}{v_0^{10}(\lambda + a_0 - 1)^{10}}.$$

The denominator is a square, and all numerator factors are the selected primes, each used once in that detector. The source factor γ^2, $\gamma = 408/409$, and the squared reciprocal trace do not change valuation parity. Hence the pole input has square reduced denominator. At preparation primes, choose λ to dominate both 1 and T p-adically; then R_j is arbitrarily close

to one, q_j is small and the inverse trace is close to two. The same γ gives local squareclasses and depths $4, 2, 2$ at $2, 3, 17$. In the real compact neighborhoods, as λ tends to infinity,

$$R_j = \prod_{\alpha}(1 - \alpha/(t_j - 1)^2) \longrightarrow 1, \quad q_j > 0, \quad W_j \longrightarrow 1, \quad b_j \longrightarrow \gamma^2/4 \in (0, 17).$$

An arbitrarily large fixed lower bound on the slope supplies all these conditions. The prepared nonzero pole-realization premise now supplies independent witnesses for the two actual pole inputs.

Finally fill the new contact outputs by (5), and take $t_1 = \lambda + T$. The unchanged rows hold by the source physical reconstruction, while the nine changed rows hold by construction. This proves completeness for every integer input and every $N \geq N_C$. Taking the maximum with the soundness threshold proves Theorem 1.1. No numerical sample tuple is used to replace this universal existence argument.

6 Balanced coefficients and exact certificates

Set $A_N = 149C^2$, $C = 149^N$. In the new unbalanced system rescale only the coordinates with parent-214 IDs 117 and 118 by $x_i = Cy_i$. All other coordinates are fixed. Divide each transformed row by its smallest common power of C and normalize to a primitive integer row. Every resulting row remains quadratic and is at most linear in C. Since $C > 0$, the change is invertible and preserves every rational solution and the input. Unlike the earlier 214 balancing, no contact coordinate needs this rescaling.

The unbalanced quartic has 2909 parameter-separated terms; the balanced one has 2910. Both have 2908 coordinate monomials for every positive coefficient parameter. The only mixed balanced coefficient profiles are $1 + 22201C^2$ and $1 + 2C^2$; no cancellation can alter the support. An independent expansion identifies a coefficient equal to H_0 and one equal to K_0C^2. Every other profile is either ℓC, with $\ell^2 \leq H_0K_0$, or $a + bC^2$, with

$$|a|K_0 + |b|H_0 \leq H_0K_0.$$

The latter bounds the profile by $\max(H_0, K_0C^2)$ on both sides of $C^2 = H_0/K_0$; the former follows by the geometric mean bound. Thus the exact height is the asserted maximum. Exact integer comparison gives $K_0149^{112} < H_0 \leq K_0149^{114}$, proving the coefficient threshold 57. The unbalanced certificate similarly identifies the unique $A_N^2 + 1$ profile and verifies threshold 53.

The directory `computation/quartic/additive212_v25` contains both complete quadratic and quartic tables, their builder, and a checker importing no producer. The checker re-expands all 188 source rows and matches every source coefficient, reconstructs all nine changed equations independently, compares all 177 retained rows, checks both physical contact normalizations and the boundary $t = 1$, and expands every new coefficient. It also verifies the two-coordinate rescaling on every row, the exact height profiles for all positive parameters, and deliberate rejection of a changed coefficient and a missing row. These finite checks do not prove the source arithmetic premises or complete an independent non-author review.

7 Source and research status

The new contribution is the additive comparison, its simultaneous realization, and its compatibility with both shorter contact circuits. The earlier attempt to delete these two outputs while preserving the multiplicative, rescaled argument did not establish this construction. Changing to (2) is what makes the unscaled second contact circuit quadratic. This is not a fixed-witness bijection with the original 214 system; soundness and completeness have been proved again

at the changed arguments. The coefficient balancing alone is an exact fixed-input bijection between the two new presentations.

The 214 source is retained as a separately attributed predecessor, not erased by its improvement. Its exact upstream commit, tables, reviews and open audit limitations remain distinct from these new proofs. In particular the full prime-pattern, pole-realization and modular-height foundations are not declared exhaustively audited by this paper. All results here have precisely the arithmetic scope stated at the beginning.

References

[1] Pinned 214-variable construction and complete Version 17 manuscript, upstream commit `1f7e1f1fb0438feda3c9f3da9cc16982126aeeed`; `research/quartic214/proofs/uniform-existence.md`, P03–P20, and `manuscripts/quartic_v17.tex`, Sections A43–A47. The numeric source and provenance are retained in the present repository.

[2] Companion paper, `papers/quartic_existence/quartic226_20261110_v21.tex`, the inverse-weight contact calculation.

[3] Companion paper, *Coefficient balancing for a 214-variable quartic*, `papers/quartic/coefficient_balancing_20261113_v24.tex`.

[4] Companion paper, `papers/local_tests/height_precision_20261113_v23.tex`, for the sharper optional exponent 2103/2048 in place of 7/6.